Python for Data Science & Machine Learning Masterclass

Comprehensive Developer Masterclass & Practical Guide

DATA SCIENCE

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Chapter 1: Data Wrangling and High Performance Pandas

Vectorization, memory optimization, and cleaning pipelines

Data preparation accounts for over 70% of a data scientist's workflow. Leveraging vectorized operations in Pandas instead of Python iterrows() leads to 100x performance improvements.

Key Takeaways & Best Practices:

- Downcast numeric datatypes to uint8/float32 to minimize memory usage.
- Use category dtype for low-cardinality string columns.